

Gary Peng

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RESEARCH INTERESTS

I am broadly interested in algorithms and complexity, and I am currently working on **batch-dynamic parallel graph algorithms**.

EDUCATION

Massachusetts Institute of Technology

Fall 2026 - Present

Doctor of Philosophy, Electrical Engineering and Computer Science

Advisor: Julian Shun

University of Maryland

Fall 2022 – Spring 2026

Bachelor of Science, Computer Science

Bachelor of Science, Mathematics

GPA: 4.00 / 4.00

RESEARCH EXPERIENCE

Rutgers University

Spring 2025 - Present

Advisor: Karthik C.S.

- Proved fine-grained hardness-of-approximation results for clique and biclique
- Discovered a simple proof (bypassing the highly non-trivial PCP Theorem) that any constant-approximation for clique requires $n^{\Omega(\log n)}$ time under the Exponential Time Hypothesis

University of Maryland

Spring 2024 – Summer 2026

Advisors: MohammadTaghi Hajiaghayi, Suho Shin

- Showed that the approximation ratio to the first-best gains-from-trade can be arbitrarily close to zero in bilateral trade with a profit-maximizing broker
- Proved that a simple single-sample mechanism simultaneously achieves constant-factor approximations to the first-best gains-from-trade, first-best social welfare, and optimal profit under certain regimes of bilateral trade

INTERNSHIPS

DIMACS Research Experience for Undergraduates, Rutgers University

Summer 2025

Advisor: Karthik C.S.

Director's Summer Program, National Security Agency (*regretfully declined*)

Summer 2025

AWARDS & HONORS

NSF Graduate Research Fellowship

April 2026

National Science Foundation

John D. Gannon Scholarship

July 2025

Department of Computer Science, University of Maryland

Award Amount: \$3,243.04

Philip Merrill Presidential Scholar

May 2025

University of Maryland

Abramowitz Award

May 2025

Department of Mathematics, University of Maryland

Award Amount: \$1,000.00

Honorable Mention for Outstanding Undergraduate Researcher Award

January 2025

Computing Research Association

President's Scholarship

February 2022

University of Maryland

Award Amount: \$12,000.00 (annually)

Platinum Contestant

April 2021

USA Computing Olympiad

ACCEPTED PAPERS

Single-Sample Bilateral Trade with a Broker

WWW26

MohammadTaghi Hajiaghayi, Gary Peng, Suho Shin

Gains-from-Trade in Bilateral Trade with a Broker

SODA25

Ilya Hajiaghayi, MohammadTaghi Hajiaghayi, Gary Peng, Suho Shin

PROFESSIONAL SERVICE

External Reviewer

WINE26, SODA26

TEACHING EXPERIENCE

Calculus II (MATH141), University of Maryland

Spring 2025

Strauss Teaching Assistant

Calculus I (MATH140), University of Maryland

Fall 2024

Strauss Teaching Assistant

Design and Analysis of Computer Algorithms (CMSC451), University of Maryland

Summer 2024

Teaching Assistant

Discrete Structures (CMSC250), University of Maryland

Spring 2023

Teaching Assistant